पदार्थ विज्ञान एवम् अभियांत्रिकी विभाग
भारतीय प्रौद्योगिकी संस्थान कानपुर

Department of Materials Science and Engineering
Indian Institute of Technology Kanpur

PLACEMENT BROCHURE

2025-2026

GET IN TOUCH

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www.linkedin.com/in/mse-iit-kanpur-18236516b/
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TABLE OF CONTENTS

Head of the Department's Open Letter

Dear Recruiter...

Materials Science and Engineering (MSE) department presents students, who are well-equipped with knowledge, skills and values essential for excelling in ever-evolving professional landscape.

MSE department fosters an environment of excellence, commitment and holistic development. We strongly believe in equipping our students with not only theoretical knowledge but also exposing practical/laboratory skills that are crucial in today's dynamic and competitive world.

MSE students are well-prepared to tackle real-world challenges undergoing rigorous curriculum, hands-on projects, internships, and industry interactions, so as to meaningfully contribute to your esteemed organizations. We take pride in the fact that MSE students excel in wide spectrum and are eager to step into industries, like automotive, aerospace, materials processing, iron and steel making, non-ferrous metallurgy, ceramics, healthcare, semiconductors, data analytics including AI/ML, and many more.

While 352 of our undergraduates of our department are trained to acquire basic understanding of the materials and quantitative analysis so that they can take up a wide variety of problem-solving challenges offered by the recruiter, our research cadre students, 83 M.Tech students, 06 BT-MT students and 121 PhDs, are provided specialized training to take up any scientific and technical challenges for the development and utilization of new processes and materials for key projects in your futuristic strive to become global leaders. With a strong foundation in mathematical and numerical methods, MSE students are also eager and capable of contributing to building software for the upliftment and knowledge creation in materials world. Beyond academics, we emphasize the importance of soft skills, leadership qualities, civic responsibility, integrity and ethical values. Our students participate in various extracurricular activities, community service initiatives, and industry interactions, which contribute to their overall personality development and prepare them to be responsible global citizens.

By recruiting from Department of Materials Science and Engineering at IIT Kanpur, you will be investing in individuals who are not only technically proficient but also possess the drive and determination to make meaningful contributions to your organization. We sincerely wish that some of our students will have the privilege of becoming part of your esteemed organization. Be assured that MSE students at IIT Kanpur are nothing short of being remarkable, adding substantial value to your esteemed institution.

MSE department wishes to build a strong relation through this recruitment and extends a sincere '*thank you*' for considering our students for various opportunities within your honoured organization. We look forward to establishing a mutually beneficial partnership and contributing constructively to your admired success.

Prof. Kantesh Balani
Head, Materials Science and Engineering
IIT Kanpur

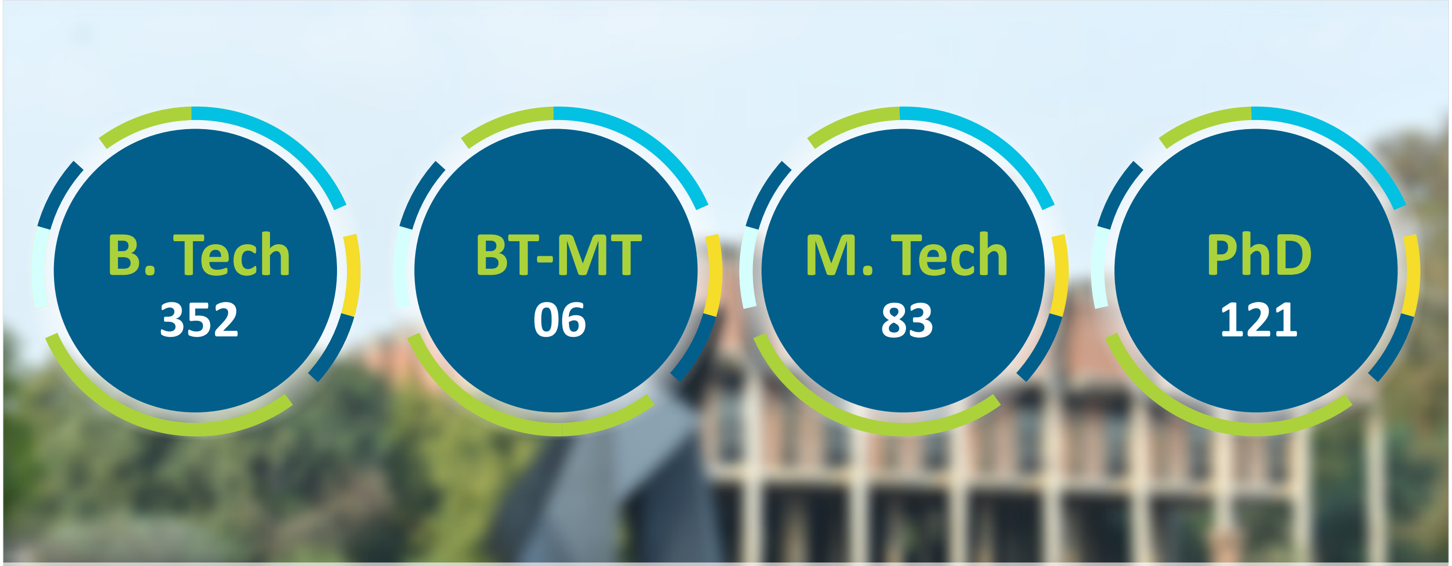
About Us

The ***Department of Materials Science and Engineering*** at IIT Kanpur with its 60 long years of legacy of excellence in undergraduate and post graduate teaching and research strives to prepare technologists and engineers to develop new materials and processes for applications in industries of metal and mining, automotive, chemical, aviation, plastic, biotechnology, semiconductor, solar and energy sectors.

The Department was established in 1960. Founded in 1960, the ***Department of Materials Science & Engineering*** at IIT Kanpur holds a prominent position among the nation's leading schools in Materials Science and Engineering. Initially named ***Metallurgical Engineering***, the department introduced its postgraduate program in 1964. Over the years, it underwent a series of adaptions as per the need of the country and society, changing from **Metallurgical Engineering** to **Materials and Metallurgical Engineering** in 1993 and eventually transitioning to the Department of Materials Science and Engineering in 2009. Since its inception, it has had a strong impact in providing knowledgeable manpower to meet the nation's demand in traditional metallurgy. The department consciously nurtures overall professional growth of the students, apart from giving quality education. The department has constantly reinvented to keep the IIT curriculum in pace with the state-of-the-art technologies.

The field of study in the department now encompasses the entire spectrum of *extractive metallurgy*, *physical metallurgy, manufacturing processes, electronics and semiconductor materials, mechanical behavior of materials, powder metallurgy, process modeling, material degradation, nanomaterials, biomaterials, ceramics, composites, recycling & material recovery and computational materials engineering (AI/ML), microstructure, materials processing, physical & mechanical metallurgy, coatings & materials for damage tolerance for fatigue, creep, fracture*, etc. This department has pioneered a unified approach for teaching and research, which has enabled us to evolve into an interdisciplinary field contributing to diverse applications and technological development.

Academic Programs



B. Tech
352

BT-MT
06

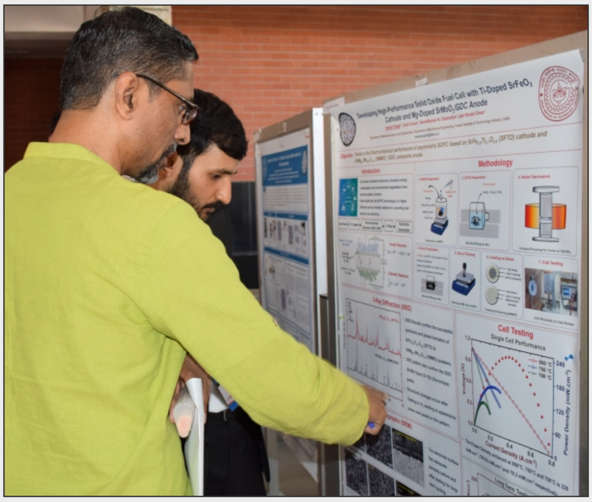
M. Tech
83

PhD
121

MSE Department at IIT Kanpur puts emphasis on enhancing the technical expertise of the students and provides hands-on Industrial exposure to students with internships, projects, and visits.

UG Coursework

- Basic Sciences
- Core Laboratories
- Departmental Courses
- Internship (2nd /3rd year)
- B.Tech Thesis
- Internship



PG Coursework

- Structure and Characterization of Materials
- Transport Phenomena
- Thermodynamics
- Mathematical and Computational Methods
- M.Tech/Ph.D Thesis
- Teaching Assistantship
- Internship

Research Facilities

MSE department at IIT Kanpur is furnished with ***world-class research facilities*** assisting students in learning vital skills and gain hands-on experience of the latest technologies used in industries and academia. Apart from the various testing and characterization laboratories, the department also houses befitting computational and modeling facilities in steelmaking, fluid dynamics, and solidification processing. We have the following labs in our department: https://iitk.ac.in/mse/MSE-Facilities/

Microstructure Characterization Facility

Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM)
Transmission Electron Microscopy (TEM), Brunauer-Emmett-Teller (BET)
Electron Probe Micro Analyzer (EPMA), Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC), X-Ray Diffraction (XRD),
X-Ray Photoelectron Spectroscopy (XPS)

Physical Metallurgy and Engineering Metallurgy Lab

Optical Microscopy
Microwave Sintering Furnace
Rolling Mill
Welding
Brazing

Electronic Materials and Thin Film Processing Lab

Pulse Laser Deposition
Electron-Beam Evaporation
Photolithography
Clean Room
Sputtering

Material Testing Lab

Universal Testing Machine
Fatigue Testing
Creep Testing
Impact Testing
Hardness Testing
Microhardness Testing

Research Centres

SAMTEL Centre for Display Technologies

To conduct R&D so as to nurture and support the growth of science and technology of electronic displays and to establish a tripartite relationship between industry, academia and governmental agencies.

National Centre for Flexible Electronics

It acts as a nodal point in India to bring academia, industry and public research organizations under one umbrella for research and development of large area flexible electronics.

Industry Partners: Applied Materials, Manipal Technologies, Chain Electronics, Mathura Manufacturing

Advanced Centre for Materials Science

Advanced Centre for Materials Science was created in 1978 with a view to make available major materials preparation and characterization facilities under one-roof. These state-of-the-art research facilities are regularly upgraded, and maintained by suitably trained competent staff.

ICME NATIONAL HUB
AN IITK-TCS JOINT INITIATIVE

Integrated Computational Materials Engineering

Integrated Computational Materials Engineering is a National Hub at IIT Kanpur - A Joint IITK-TCS Initiative

Centre for Nanosciences

The centre is aimed to provide various nanomission & nanotechnology related fabrication & characterziation tools for fundamental research support to startups and other academic & industrial partners.

Seminar/Talks (2024-2025)

https://iitk.ac.in/mse/conference-seminar.php

Dr. Ajay Gautam, R&D Scientist at Samsung SDI Europe GmbH in Germany, delivered a talk on "***Designing lithium argyrodite-based solid electrolytes for next-generation lithium-ion batteries***" on May 27, 2025 .

Dr. Sudipta Pramanik, Assistant Professor, School of Minerals, Metallurgical and Materials Engineering, IIT Bhubaneswar, delivered a talk on "***Investigation of Additively Manufactured Materials***" on Apr. 02, 2025.

Dr. Sudhanshu Sharma, Department of Chemistry, Indian Institute of Technology Gandhinagar, Gandhinagar, India, delivered a talk on "***Heterogeneous Catalysis Using Metal Oxides And High Entropy Alloys***" on Nov. 26, 2024 .

Dr. Ing Kong, Associate Professor in the Department of Engineering at La Trobe University, Australia, delivered a talk on "***Polymer Nanocomposite Bioink For 3D Bioprinting And Tissue Engineering Applications***" on Nov. 19, 2024.

Dr. Brahma Deo, MGM Chair Professor, IIT Bhubaneswar Coordinator, Blended-Mode M.Tech. Program on Advanced Maintenance Technology, delivered a talk on "***Dynamic Vibration Analysis For Advance Failure Prediction Of A Rotating Burner***" on Nov. 11, 2024.

Dr. Karthik Akkiraju, Postdoctoral Associate, The Yale School of the Environment, Yale University, delivered a talk on "***From The Atom To The Household: Sustainable Materials Design For Human Development***" on Sep. 27, 2024.

Dr. Bhargavi Rani Anne, Assistant Professor in Department of Metallurgical and Materials Engineering, National Institute of Technology, Raipur, delivered a talk on "***Thermally Activated Deformation Mechanisms In Ti-6al-4v***" on Jul. 24, 2024.

Prof. Arvind Agarwal, Distinguished University Professor in the Department of Mechanical and Materials Engineering at Florida International University (FIU), Miami, FL, USA, delivered a talk on "***Multifunctional Coatings And Composite Materials For Lunar And Space Missions***" on Jul. 19, 2024.

Dr. Sanghamitra Moharana, Assistant Professor in the Department of Metallurgical and Materials Engineering, National Institute of Technology Warangal (NITW), delivered a talk on "***Investigating Lithium-ion Battery Degradation And Its Mitigation***" on Jul. 16, 2024.

Dr. Venkatraman Gopalan, Department of Materials Science and Engineering, and Department of Physics, Pennsylvania State University, University Park, PA, delivered a talk on "***Light-matter Interactions In Electronic Oxides***" on Jul. 09, 2024.

Dr. Heena Khanchandani, Post-doctoral Research Scientist, Department of Materials Science and Engineering at Norwegian University of Science and Technology in Trondheim, Norway, delivered a talk on "***Atomic scale investigation of hydrogen embrittlement in austenitic steels***" on Jun. 26, 2024 .

Dr. Swastika Banthia, Researcher, Surface Engineering Research Group, R&D division of Tata Steel Limited, Jamshedpur, delivered a talk on "***Functionally Graded Materials: A Strategic and Sustainable Approach for the Future***" on Jun. 24, 2024.

Dr. Ameey Anupam, Research Associate at a DST-funded Centre of Excellence for thermal spray coatings at IIT Ropar, delivered a talk on "***High Entropy Alloys and their Thermal Spray Coatings for Critical Applications***" on Jun. 10, 2024.

Dr. Swathi Ippili, Contract Professor at Chungnam National University, Republic of Korea, delivered a talk on "***Innovations in Nanostructured Materials for Sustainable Energy Applications towards Realizing Self-Charging Power Units***" on Jun. 06, 2024.

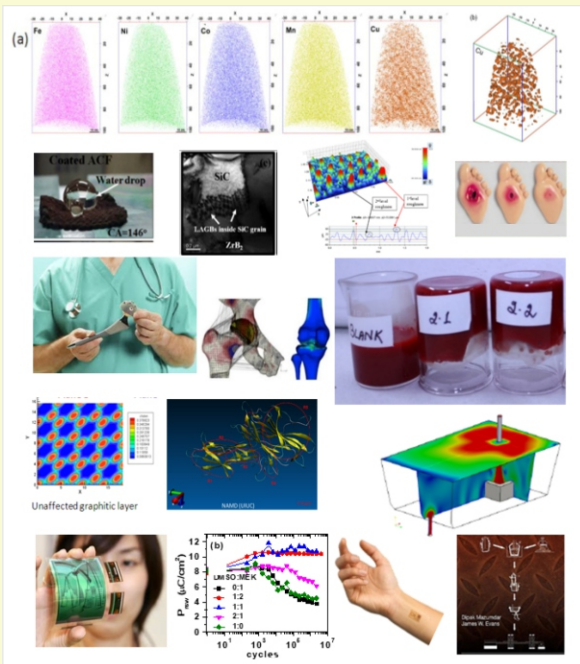
Research Themes and Areas

Research Themes

- **Health Care/Biomedical**
- **Energy and Environment**
- **Electronic Materials & Devices**
- **Railways, Automobiles, Space and Defence Technologies**
- **Iron, steel & other metals**

Research Areas

- Semiconductor Package
- Computational Materials Science
- Electroceramics
- Iron and Steel Making
- Flexible and Organic Electronics
- Manufacturing Processes
- Material Degradation
- Mechanical Behavior of Materials
- Nanomaterials and Nanotechnology
- Physical Metallurgy
- Powder Metallurgy
- Process Modeling
- Recycling and Material Recovery
- Structural Ceramics
- Biomaterials



Projects and Collaborations

Areas of Ongoing Projects

- Sustainable Iron and Steelmaking, Process Modelling
- Flexible electronics, materials and devices, semiconductor materials, Organic Electronics
- Computational Materials Science, Finite Element Method, Integrated Computational Materials Engineering
- Physical Metallurgy, Phase Transformation
- Environmental degradation of alloys
- Biomaterials, Protein Patterning
- Multi-component Diffusion, Thermodynamics
- Powder Metallurgy, Ceramic Processing, Sintering, Solid Oxide Fuel Cells, Batteries
- Grain Boundary Engineering, Severe Deformation Processing
- Mechanical Behaviour of Materials
- Stereology, Crystallography
- Glassy Alloys, Quasicrystals
- Nanomaterials/ Composites
- 3D and additive Manufacturing
- Recycling and material Recovery

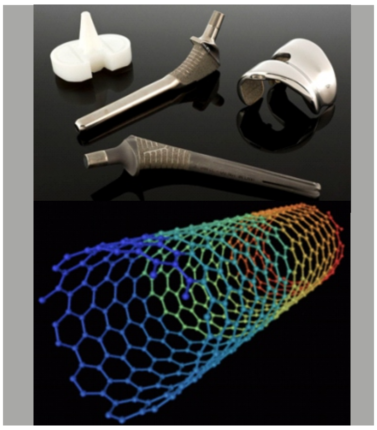
Collaborations

Academic Curriculum

Metallurgical Engineering

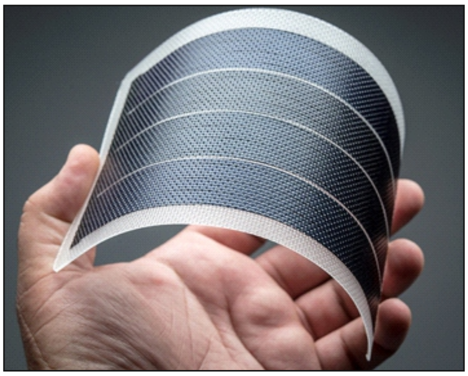
- Iron and Steel Making
- Principles of Metal Extraction and Refining
- Phase Transformations
- Thermodynamics & Phase Equilibria
- Mechanical Behaviour of Materials
- Materials Failure: Analysis and Prevention
- Structure and Characterization of Materials
- Diffusion in Solids
- Materials Recyling

Nanomaterials and Biomaterials



- Introduction to Biomaterials
- Materials Science Technologies for Applications in Life Sciences
- Transmission Electron Microscope Nano Analysis of Materials
- Nanostructures and Nanomaterials
- Characterization and Properties
- Coatings

Electronic Materials



- Electronic Devices and Characterization
- Electro-ceramic Materials and Applications
- Computer Simulations in Materials Science
- Technology of Thin Films and Device Fabrication
- Introduction to Semiconductor Packaging
- Energy Materials and Technologies
- Materials for Semiconductor Industry
- Display Technology

Materials in Manufacturing

- Materials Processing
- Selection & Design of Engineering Materials
- Manufacturing Processes
- Solidification Processing
- Heat Treatment and Surface Hardening
- Powder Metallurgy
- Introduction to Lightweight Alloys

Departmental Activities

Deparment Bodies

Indian Institute of Metals - Kanpur chapter organizes Materials Quiz workshops and conferences, involving student-faculty interaction

Materials Society(MatSoc) is an integral student community which organizes various departmental talks, workshops, events and recreational activities

Material Advantage @ IIT Kanpur is a window providing access to the materials professional's most eminent societies like ASM, TMS, AIST and Acers.

AWARD FOR MOST RECRUITED STUDENTS (2024-2025)

Material Advantage @ IIT Kanpur, the student technical chapter of the Materials Science and Engineering Department, received "**Most Recruited Students**" during **the 2024 Fall Membership Challenge**, which includes a citation and a cash award of US$1000.

MATERIALS CAMP


8-11 May 2025

Material Advantage @ IIT Kanpur has organized a "Materials Camp" at IIT Kanpur in collaboration with the ASM International Kanpur Chapter, the Indian IIM Kanpur Chapter, the Indian National Academy of Engineering (INAE) Kanpur Chapter, and MATSOC. The "Materials Camp" attracted 42 students and nine teachers from different schools in Kanpur.



Deparment Bodies

MATERIALS SCIENCE SPORTS LEAGUE

The year began with a warm and informative orientation session, aimed at welcoming new students and helping them integrate smoothly into the academic and cultural fabric of the department. This was followed by the much-anticipated sports league, which brought together students, faculty, and staff in a spirited display of camaraderie and healthy competition across multiple sports.


January 2025

SURGE SESSION

12 January 2025

MATSOC conducted a session on SURGE where Y22 MSE students shared their experience. The Objectives, Prerequisites, Formalities, and Benefits of taking part in SURGE were discussed. The session emphasized finding mentors and highlighted the prospects of this opportunity. Participants gained insights to enhance their chances of selection and understand the potential advantages of participating in the program. The session ended with an interactive Q&A time.

INTERNSHIP SESSION

MatSoc conducted an interactive event titled "Internship Session," where Y22 MSE students shared their experiences from interviews across a range of profiles, including MSE Core, Software Development (SDE), Analytics, Finance, and Consulting. They provided valuable insights on resume building, available opportunities, and strategic preparation, helping participants better understand the internship landscape and how to stand out in the application process.
The session also featured a special segment where Prof. Anish Upadhyay discussed core opportunities in materials science and engineering, offering students a clear perspective on academic and industrial pathways within the core domain.

NSRS-2025

The National Symposium of Research Scholar on Metallurgy and Materials (NSRS 2025) was organized by the Department of Materials Science and Engineering, Indian Institute of Technology Kanpur on **28th March to 2nd February 2025**. It was jointly supported by the Indian Institute of Metals (IIM) Kanpur Chapter, Material Advantage IIT Kanpur and MATSOC The Materials Society. Envisioned to raise the culture of research and collaboration amongst the research scholars of the country, NSRS-2025 has been a successful symposium that has seen extensive support and participation.

Prof. Ravi Kumar N.V. (IIT Madras) delivered an engaging talk on the synthesis of advanced ceramics via a precursor route, highlighting their potential for high-temperature applications. Prof. Rajiv Shekhar (IIT Kanpur) concluded the academic sessions by urging researchers to embrace an entrepreneurial mindset, emphasizing that impactful research should go beyond academia to address pressing technological needs.

NSRS 2025 showcased a wide spectrum of research through 56 oral and 26 poster presentations, covering themes such as energy materials, high-entropy alloys, functional materials, and computational modeling. The symposium also featured insightful industry-oriented sessions. Dr. Ravi Bhatkal, Managing Director of MacDermid Alpha Electronic Solutions, and Dr. Alok Gupta of Alok Gupta & Associates, delivered talks on advanced materials for electronics manufacturing and patent filing essentials for metallurgists, respectively. Their contributions highlighted opportunities for practical implementation of academic research.

A dynamic panel discussion involving faculty members and research scholars addressed emerging challenges in metallurgy and materials science, encouraging meaningful exchanges on future research directions. Beyond technical sessions, the symposium included cultural programs and a networking dinner, which fostered collaborations and strengthened the community spirit.

The event was generously supported by industry sponsors, including MacDermid Alpha Electronic Solutions, Alok Gupta & Associates, Chennai Metco, Frontier Steels, Macro Systems, RR&Co, and Thermo-Calc, underscoring the importance of academia–industry collaboration.

The NSRS series aims to foster impactful research and collaboration among scholars across India.

Prof. Vivek Verma, MSE IIT Kanpur has created a ***biodegradable haemostatic sponge*** from ***red seagrass*** and ***cellulose*** that stops bleeding in under a minute. Aimed at emergency use—like road accidents and trauma—it speeds up clotting from 8 to 1 minute by quickly absorbing moisture.Tested extensively, the sponge is lightweight, compact, and ideal for first-aid kits. It holds three patents (one with DRDO, two with IIT Kanpur) and combines natural materials with advanced science. Prof. Verma noted its potential to save lives in areas with limited medical access. Human trials will begin soon.

Prof. Anish Upadhyaya has been awarded '**Pandit Girish & Sushma Rani Pathak Chair**' of IIT Kanpur from June 15, 2024, for a period of three years.

Prof. Shikhar Misra has been awarded '**Scientific High Level Visiting Fellowship**' for ~10 day research trip to France.

Prof. Krishanu Biswas has been selected to receive the **MRSI Medal for 2024**. This honor was presented to him during the MRSI Annual General Meeting, held at UGC-DAE-CSR, Indore, from December 3-6, 2024.

Faculty List

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Krishanu Biswas
Website: https://home.iitk.ac.in/~kbiswas

Niraj Chawake
Website: https://home.iitk.ac.in/~nchawake

Anshu Gaur
Website: https://home.iitk.ac.in/~agaur

Srinu Gangolu
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Nilesh Prakash Gurao
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Sarang Ingole
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Shikhar Krishn Jha
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Monica Katiyar
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Heena Khanchandani
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Kallol Mondal
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Shobit Omar
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Sandeep Sangal
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Rahul Sarkar
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Shashank Shekhar
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Sudhanshu Shekhar Singh
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Soumya Sridar
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Raghupathy Yuvraj
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Anish Upadhyaya
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Vivek Verma
Website: https://home.iitk.ac.in/~vverma

Shivam Tripathi
Website: https://home.iitk.ac.in/~shivamt

Dipak Mazumdar
(Emeritus Faculty)
Website: https://home.iitk.ac.in/~dipak/

Distinguished Alumni

Mr. Suresh Pandey
(BT/MME/1965)
Former Director,
Bokaro Steel Plant
(Management excellence)

Prof. Raj N. Singh
(BT/MME/1967)
Regents Professor,
Oklahoma State University
(Member of the National
Academy of Engineering)

Prof. Jagdish Narayan
(BT/MME/1969)
Prof., Carolina
State University
(Academic Excellence)

Mr. B. K. Shah
(BT/MME/1974)
Exec. Director, AIA
(Entrepreneurial
Excellence)

Mr. Som Mittal
(BT/MME/1973)
Former Chairman,
NASSCOM
(Management Excellence)

Mr. Anil Bansal
(BT/MME/1977)
President, Reality National
Management, Inc.
(Distinguished
Alumnus Award 2023)

Mr. Jai Shankar Sharma
(BT/MME/1977)
Mentor of the Bangalore
Chapter, IIT Kanpur
(Distinguished
Alumnus Award 2023)

Shree Pradeep Goyal
(BT/MME/1978)
(Founder Chairman and
Managing Director of
Pradeep Metals Limited,
Mumbai)

Prof. Veena Sahajwalla
(BT/MME/1986)
Scientia Professor,
UNSW
(Academic Excellence)

Dr. Pramath Raj Sinha
(BT/MME/1986)
Founder,
Ashoka University
(Service of the society
at large)

Prof. Arvind Agarwal
(BT/MT/MME/1993/1995)
(Professor, Florida
International
University (FIU), USA)

Prof. Aparna Singh
(BT/MME/2007)
Professor IIT Bombay
(Young Metallurgist
of the year)

Past Recruiters

TATA STEEL	APPLIED MATERIALS	intel	micron
JSW Steel	vedanta transforming for good	ArcelorMittal	JSL JINDAL STAINLESS
SAINT-GOBAIN	सेल SAIL	TATA MOTORS	TVS
alphonso The TV Data Company	KLA Tencor	इंडियनऑयल IndianOil	Hero
EXL	बी एच ई एल BHEL	BAJAJ	LARSEN & TOUBRO
SAMSUNG	Wrig Nanosystems Pvt. Ltd.	OLA	Reliance Industries Limited
एनटीपीसी NTPC	tok	EXIDE	HSBC
HFCL	VOLVO	UL	NOKIA
ICICI Bank	tcs TATA CONSULTANCY SERVICES	CREDIT SUISSE	Schlumberger

Past Recruiters

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